



SC1006

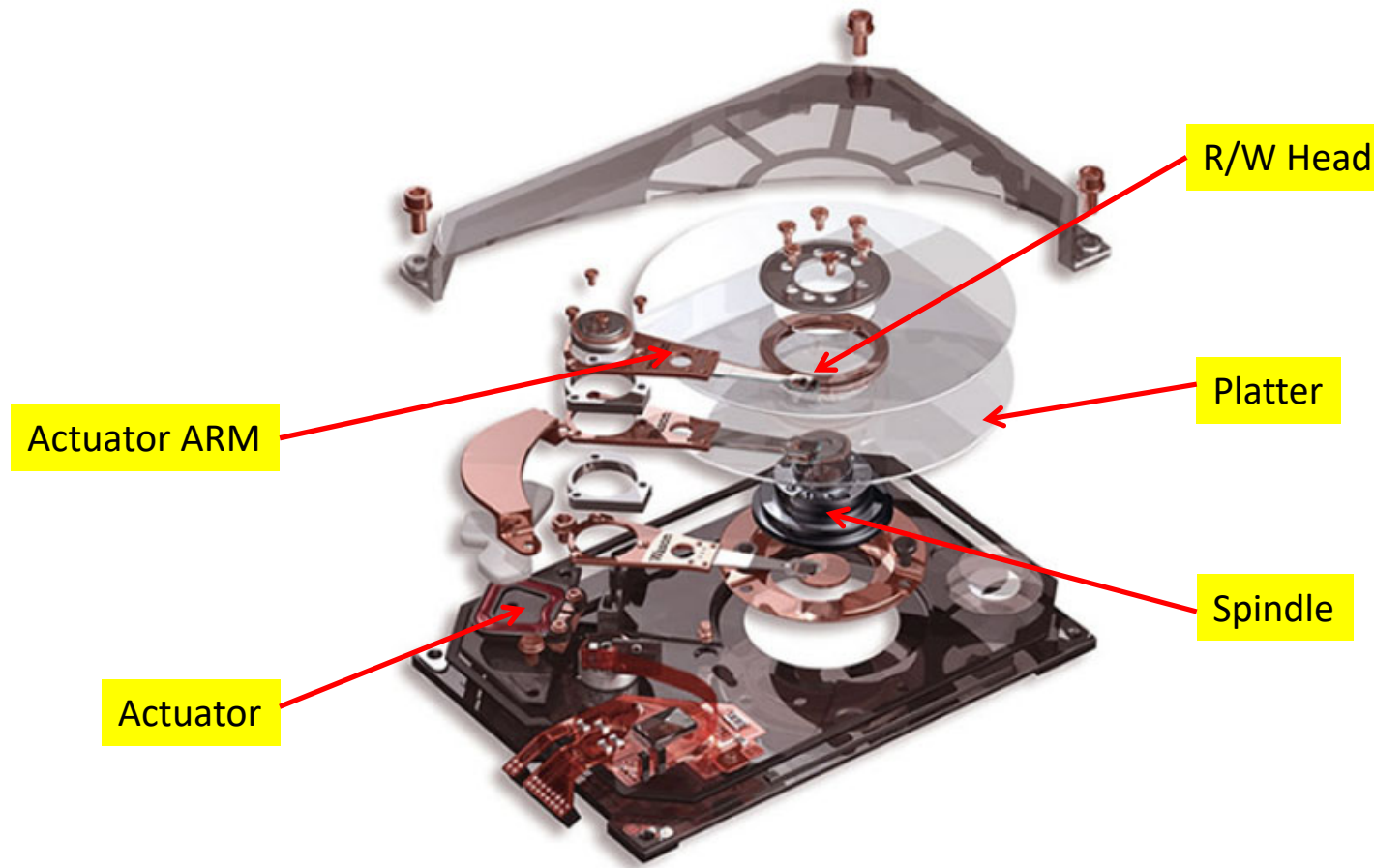
Computer Organisation and Architecture

HDD & SSD

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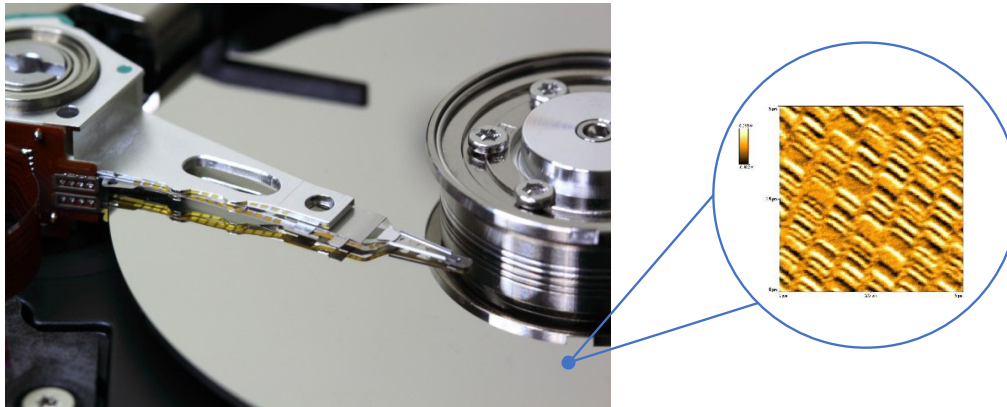


Magnetic Hard Disk

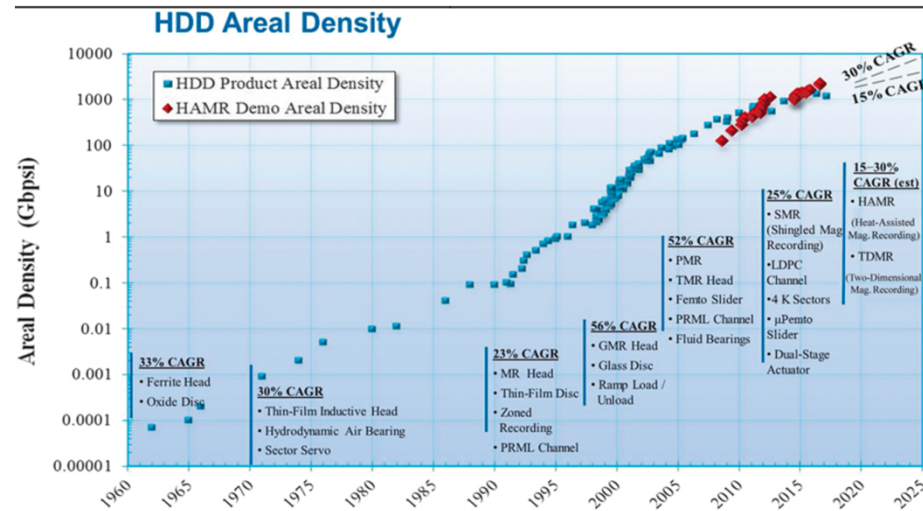


Source: <https://www.backblaze.com/blog/hdd-versus-ssd-whats-the-diff/>

Magnetic Hard Disk

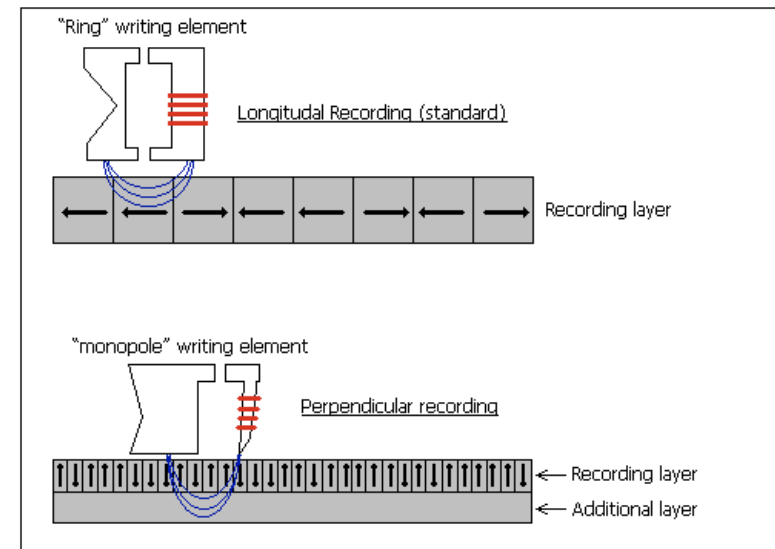


[Source] <https://www.digitaltrends.com/computing/blackblaze-hard-drive-report-seagate-western-digital/>



Source: Cambridge

Longitudinal vs Perpendicular Recording



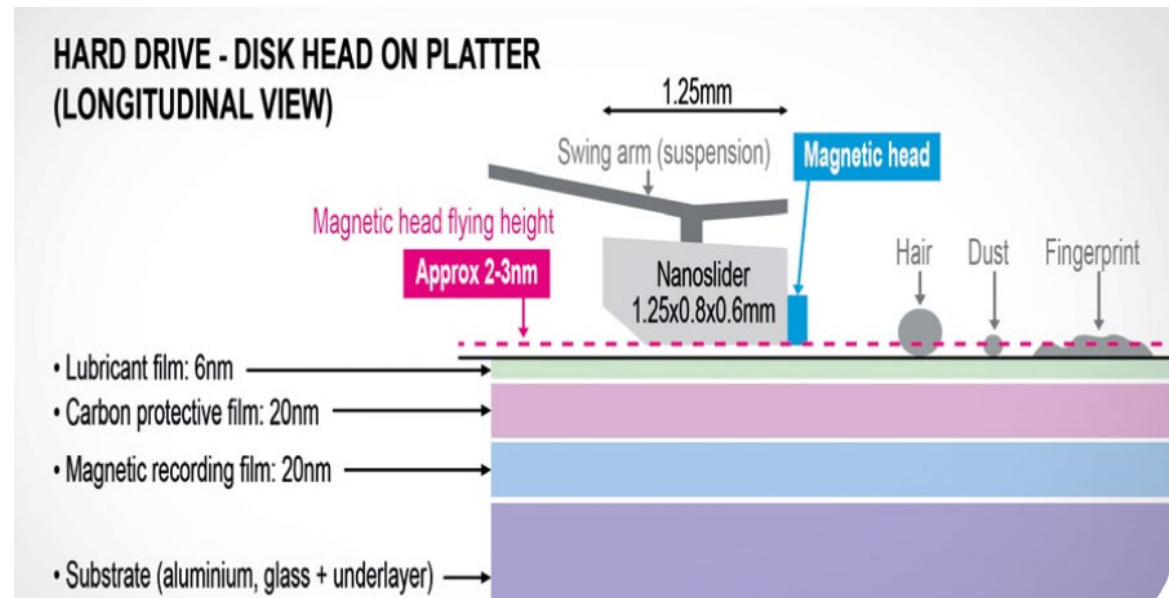
[Source] Robert Fontana et al, IBM Areal Density Comparison Paper, 2010

- Stores data by magnetizing a thin film of **ferromagnetic media** on the circular disk known as **platter**.
- Video on Introduction to Hard Disk.
<https://www.youtube.com/watch?v=kdmLv1n82U>

HDD Technology

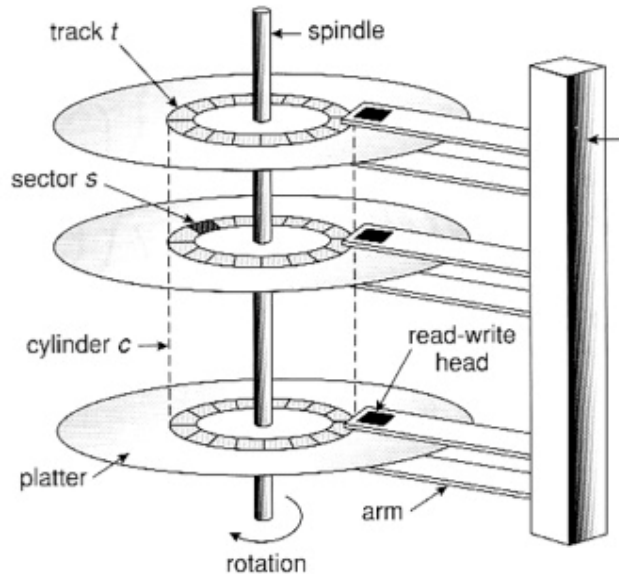


- The magnetic head uses the **wind pressure** generated by the high speed rotation of the disk to fly above the disk surface, similar to the principles employed by the aircraft.
- Fly height is lower than a finger print mark.



Source: datarecoverydublin

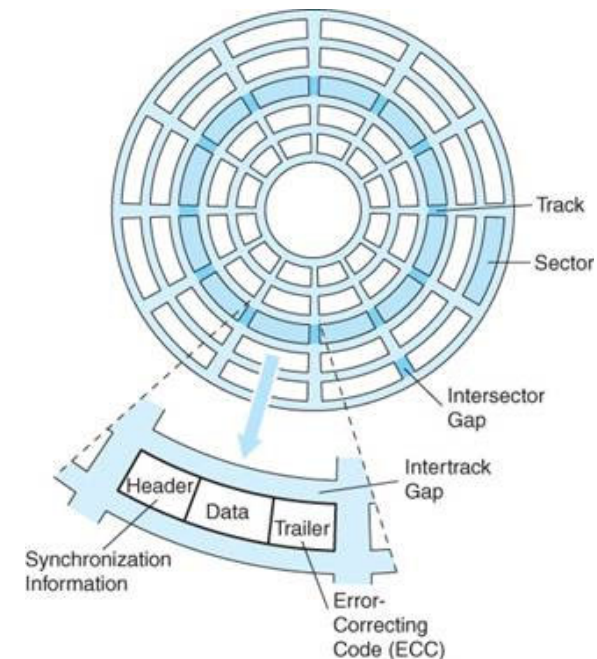
Magnetic HDD Data Organization



[Source] https://www.researchgate.net/figure/Sector-and-Platter-Design-1_fig11_303945698

- Data is stored on the surface of the platter in concentric rings called tracks.
 - Gaps between tracks to minimize interferences from adjacent tracks.
- Tracks are divided into sectors.
 - Minimum data block size is a sector.
- Stored data consists header, data and trailer.

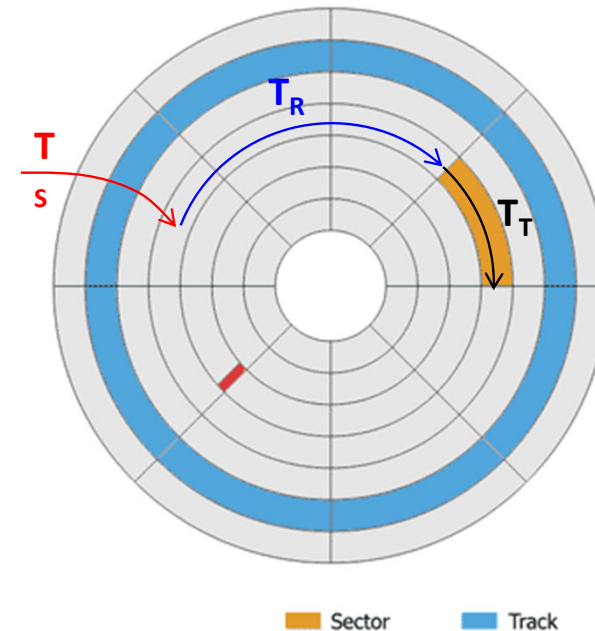
- Computers often use magnetic hard disks for large secondary storage devices.
 - One or more platters on a common spindle.
 - Platters are covered with thin magnetic film.
 - Platters rotate on spindle at constant rate.



[Source] http://www.edwardbosworth.com/My5155_Slides/Chapter12/DiskBasics.htm

HDD Transfer Rate

- Seek time (T_S)
 - Time taken for the head to move to the correct track.
- Rotational Delay (T_R)
 - Time taken for the disk to rotate until the read/write head reaches the starting position of the target sector.
- Access Time (T_A)
 - Time from request to the time the head is in position ($T_S + T_R$)
- Transfer Time (T_T)
 - Time required to transfer the required data after the head is positioned.



HDD Transfer Rate

- T_R (Rotational Delay) is dependent on the rotational speed, Revolutions Per Minute (RPM), of the disk. For calculations, RPM is usually converted to Revolutions Per Second (RPS). i.e. $RPS = RPM/60$
- For a random section, average rotational delay $T_{R,AV}$ may be calculated as

$$T_{R,AV} = \frac{0.5}{RPS} \text{ seconds}$$

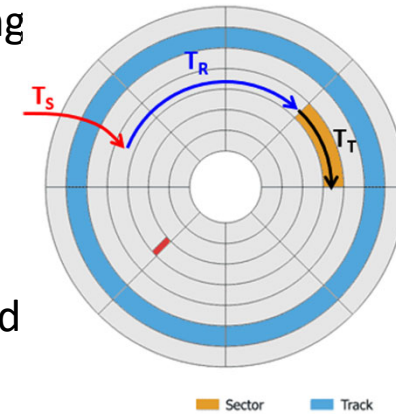
- T_T (Transfer Time) is dependent on the rotational speed of the disk, the Track Density D_T (number of sectors per track), Sector Density D_S (number of bytes per sector) and the number of bytes N for the transfer.



$$T_T = \frac{N}{RPS * D_T * D_S}$$

HDD Transfer Rate Example

- A magnetic hard disk rotates at **15000 RPM**, with the following properties:
 - Average Seek Time, $T_S = 4\text{ms}$
 - Track density, $D_T = 500$ sectors per track
 - Sector density, $D_S = 512$ bytes per sector
- Calculate the **total time** T_{TOTAL} it takes to read a **3 KB file** stored **consecutive sectors** on the same track?



[Solution]

$$\text{RPS} = 15000/60 = 250 \text{ per second}$$

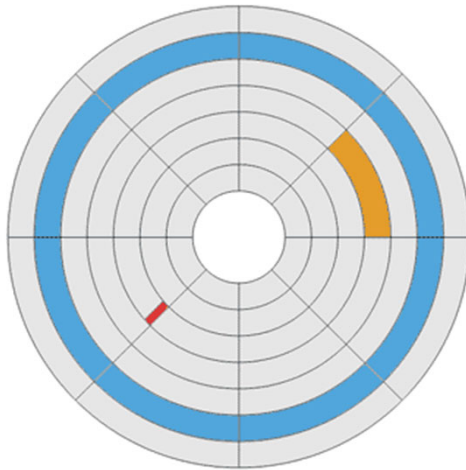
$$\text{Access time } T_A = T_S + T_R = 4 \text{ ms} + (0.5/250) = 6 \text{ ms}$$

$$\text{Transfer time } T_T = 3072/(250*500*512) = 48 \mu\text{s}$$

$$\text{Total time, } T_{\text{TOTAL}} = T_A + T_T = 6.048 \text{ ms}$$

Notice $T_A \gg T_T$. This example shows that accessing file whose data is distributed across sectors on different tracks on the magnetic hard disk would potentially incur more time.

Physical Layout

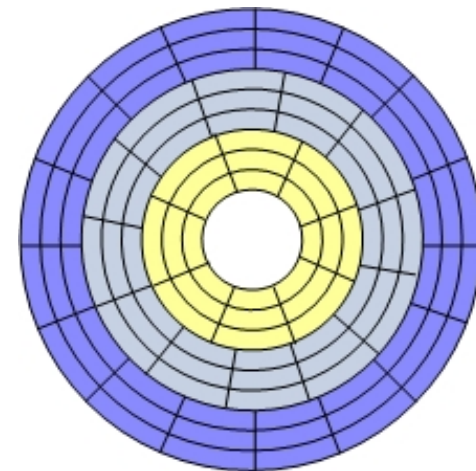


- Early HDD

- Physical Layout refers to the actual layout design on the HDD.
- Equal number of sectors per track and each sector has the same data size.
- Same-numbered Tracks from different surfaces formed a cylinder.
- The early hard disks were implemented using this topology to simplify the controller design.

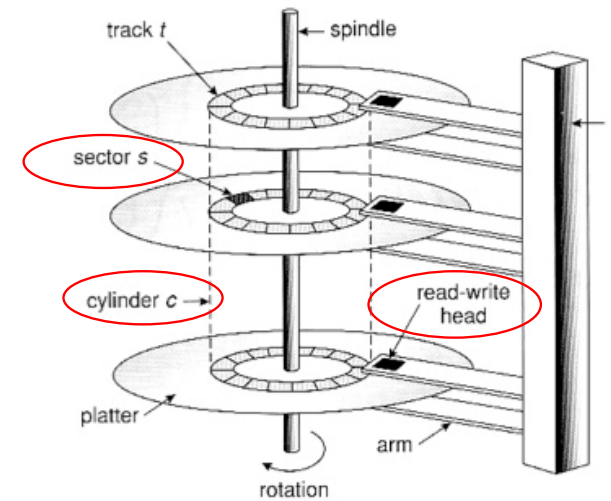
- Modern HDD

- Having equal number of sectors per track means that sectors at the outer tracks are wider.
- Waste physical space as bit density of those sectors are not optimal.
- Zone bit recording technique addresses space wastage.
- Tracks are divided into zones, with differing number of sectors per track for different zones.

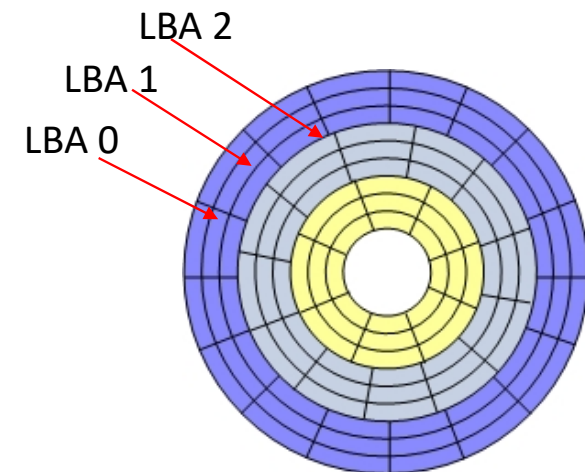


Logical Layout

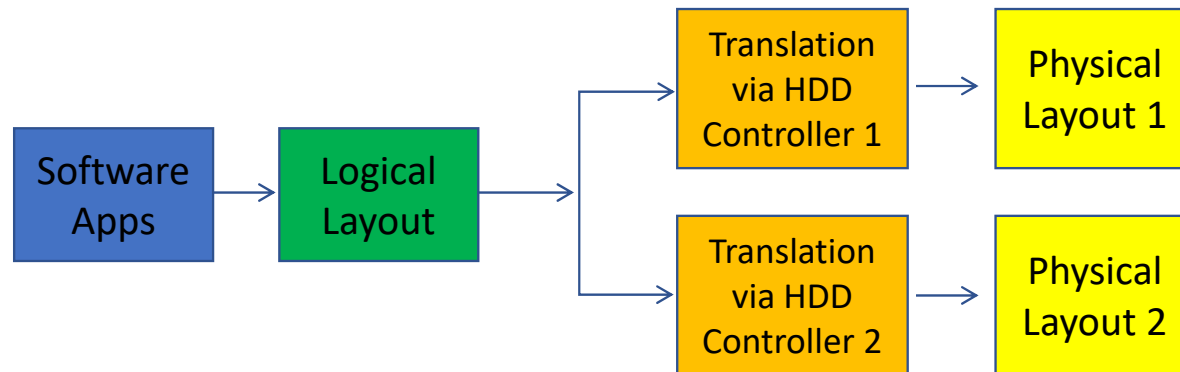
- How **software** sees & addresses HDD data?
- **Address translation**: needed to map the physical to logical locations. Two addressing scheme: **CHS** and **LBA**.
- **Cylinder-Head-Sector (CHS)**
 - Legacy scheme using the old HDD physical structure.
 - Made **obsolete** in recent standards due to addressing limitation & more **complex** formatting.
- **Logical Block Addressing (LBA)**
 - Simple **linear** addressing starting from LBA=0 as first block, LBA=1 as second block and so on.
 - 48-bit LBA standard allows addressing up to 128PByte. 1PByte=2⁵⁰ Bytes.



[Source] https://www.researchgate.net/figure/Sector-and-Platter-Design-1_fig11_303945698



Logical vs Physical Layout



- Logical layout is needed in order to provide a common interface to all software developer
- If only physical layout is available, software has to be tuned for every HDD that uses a different physical layout
- Having a standard such as LBA allows the software developer to write application that can be used in any HDD.
- HDD vendor will provide the necessary firmware in their controller to perform the logical to physical layout address translation.



Modern Day HDD

- Detail physical layout not given due to complex zone bit recording.
- Only the average transfer rate is given these days.
- Use of Cache and Buffers to speed up data transfer.
- However, concept and limitation of the magnetic HDD's physical design still valid
 - Actuator ARM is fixed and access has to be sequential, once data is missed, it has to wait for a disc revolution.
 - Seeking from track to track takes time as the R/W head needs to align to the starting sector.
 - More efficient to read in blocks rather than random access
 - Vulnerable to motion

Solid State Drive (SSD)

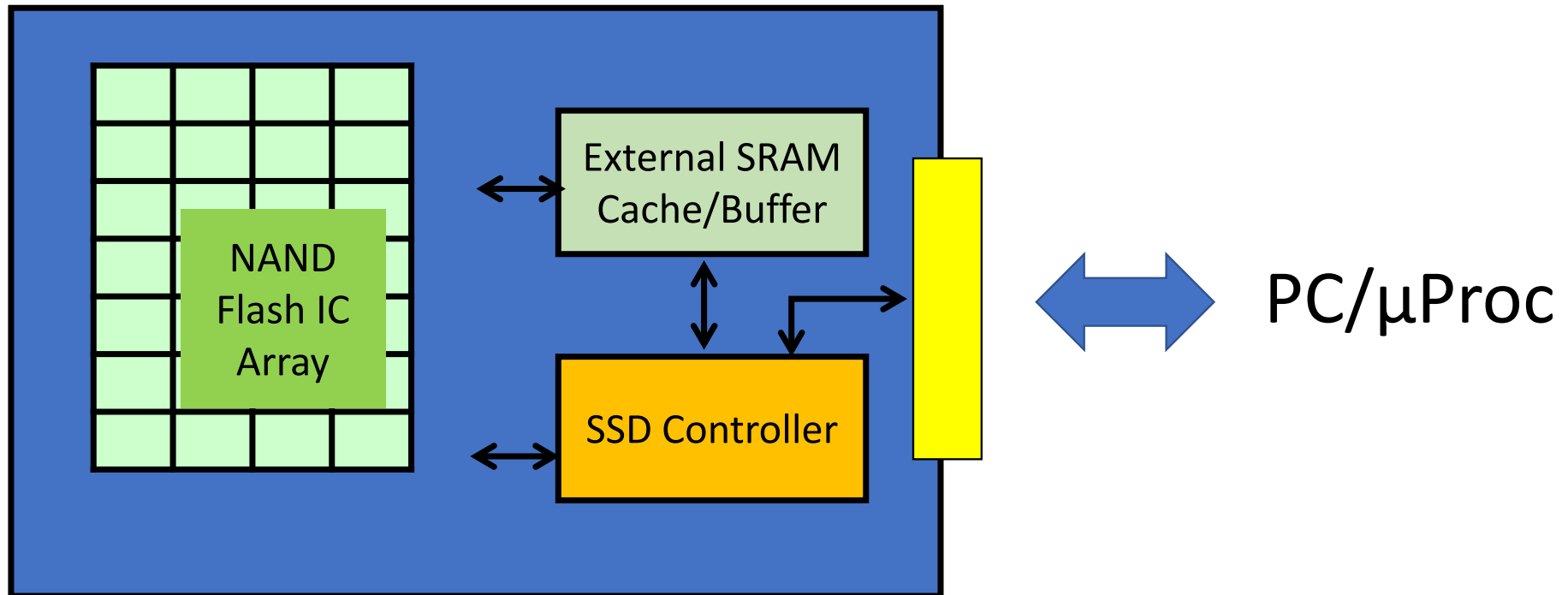
- Solid-state drives (SSD) are becoming popular
 - Memory array based on NAND-FLASH or NOR-FLASH.
 - Still more expensive than magnetic hard-disk.
- Flash memory has limited program-erase cycles (3,000 to 1,000,000).
- Various techniques used to extend the life of SSD
 - Wear levelling is a technique used to extend the life of the SSD disk by distributing data erase/write operations evenly over the entire disk.
 - Use External RAM as buffer to minimize the number of writes to Flash in SSD.
 - Error Correction Code. Ability to recover from one or more bits of error in media.
- MTBF (Mean Time To Failure) of recent SSD is comparable to that of HDD.



[Source] <https://www.kingston.com/en/blog/pc-performance/ssd-form-factors>

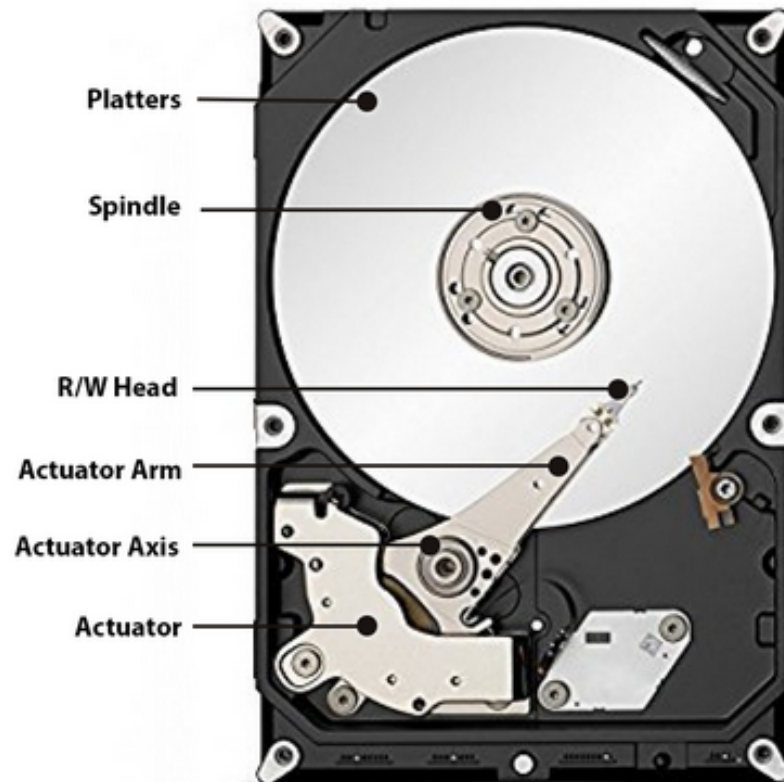
SSD System Block Diagram

Solid-State Drive



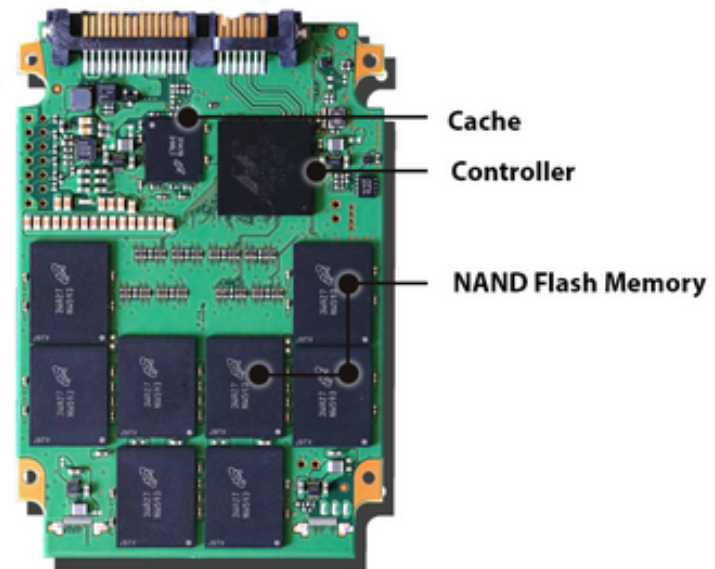
HDD vs SSD

HDD
3.5"



Shock resistant up to 55g (operating)
Shock resistant up to 350g (non-operating)

SSD
2.5"



Shock resistant up to 1500g
(operating and non-operating)

HDD vs SSD

- HDD

- Pros

- Lower Cost Per Bit,
 - Almost infinite Erasure cycles

- Cons

- Consist of Moving Mechanical Parts so more prone to crashing if HDD is dropped or shaken.
 - Heavier and larger physical profile.
 - Slower transfer rate

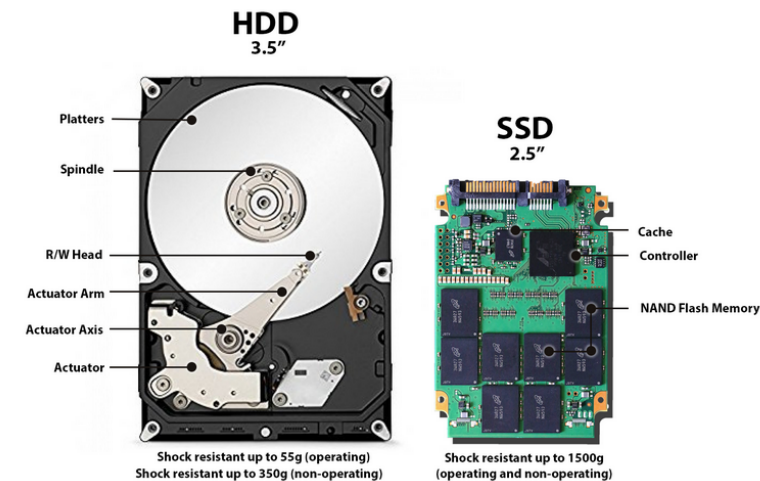
- SSD

- Pros

- No moving parts. More robust to movement.
 - Lighter and occupy less space
 - Higher Transfer rate

- Cons

- More Costly compared to HDD
 - Finite number of Erasure cycles



Source: <https://www.backblaze.com/blog/ssd-vs-hdd-future-of-storage/>

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